

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 17

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. You execute the following command in your DBMS:

```
CR vehicles 3 1
vin S
make S
year I
```

- A. List exactly what information will be added to the `catalog` relation for this new table.
- B. How many tuples will be added to the `columns` relation? List the `Attname`, `Attdomain`, and `Attproposition` for each.
- C. Why must the system prevent a user from manually calling `DE catalog`?

2. You are inserting a tuple into the `vehicles` table with the primary key `vin = "V101"`.

- A. Using the hash function for your DBMS, calculate the target bucket (data block number) for this tuple.
- B. If the target data block is already full (contains 4 tuples), describe the steps the `dbput` function must take to store this new tuple in an overflow block.
- C. What is the maximum number of blocks a single base relation can occupy?

3. You are performing a search for a specific `vin`. Your buffer currently holds 6 blocks:

- 1 Bitmap block
- 1 Catalog header
- 1 Columns header
- 3 Data blocks from a previous operation

- A. If you need to read the header block for the `vehicles` relation, which of the 9 total allowed buffer slots should it occupy?
- B. Explain the danger of not flushing a "dirty" data block to the simulated Linux disk file before reading a new block into that same buffer slot.
- C. How does the Bitmap (Block 0) help the system determine where to allocate a new block when a relation grows?

4. You run a query that creates a result called `result_table`.

- A. In the catalog, what will be the value of the `Kind` attribute for `vehicles` vs. `result_table`?
- B. Contrast the storage structure of `vehicles` (base) with `result_table` (derived).

C. Why does the system assume the entire tuple is the key for derived relations?

5. Each slot in your data blocks is 64 bytes.

A. What is the purpose of the first byte (the flag) in a slot?

B. If an attribute is a string, what is its maximum allowed length, and why?

C. Describe the two-step process required to change a key.

- **Welcome: Sean**

- [LogOut](#)

